Arnav Khandelwal

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SUMMARY

CS-AIML undergraduate turning research ideas into working systems — across computer vision, generative AI, and NLP. Comfortable with the full ML lifecycle, from feature engineering to FastAPI deployment. Looking to contribute to meaningful work in **AI/ML Engineering or Data Science**.

SKILLS

Programming Languages	Python, C, C++, SQL
Frameworks & Libraries	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, FastAPI, Hugging Face Transformers, LangChain
Tools & Platforms	Git, GitHub, Docker, Linux
Relevant Coursework	Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms, Database Management Systems

PROJECTS

Deep Learning-Based Breast Cancer Detection (BreaKHis Dataset) `GitHub`

- *Tools & Libraries: Python, PyTorch, Keras, NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV*
- Designed and trained CNN models for automated breast cancer classification using histopathology images.
- Built preprocessing pipeline with normalization and augmentation, reducing overfitting by 18%.
- Achieved 93% accuracy and 0.85 F1-score on validation data.
- Demonstrated applicability of deep learning for healthcare decision support.

AI-Based Live Screen Analyzer (GenAI + OCR) `GitHub`

- *Tools & Libraries: Python, OpenCV, NumPy, EasyOCR, MSS, psutil, Ollama (Phi-3)*
- Built a real-time AI system that captures screen activity, extracts textual information via OCR, and interprets it using a local Generative AI model (Phi-3), acting as an intelligent on-screen assistant.
- Implemented adaptive frame processing with EasyOCR and context aggregation pipeline, enabling sub-second inference latency and live system metrics (CPU, Memory, FPS).
- Integrated Ollama as a local LLM runtime with Phi-3 for privacy-preserving, offline GenAI inference with context-aware screen insights.
- Designed an interactive OpenCV-based UI with real-time OCR overlay and AI-generated insight display.

Agentic Honeypot API (AI-Based Scam Detection System) `GitHub`

- *Tools & Libraries: Python, FastAPI, Scikit-learn, NumPy, Pandas, Hugging Face Transformers*
- Built an AI-powered scam detection API that simulates victim personas and classifies fraudulent conversations using NLP models (Naive Bayes, SVM), achieving 90%+ accuracy.
- Developed an end-to-end ML pipeline using TF-IDF for real-time scam detection and threat extraction, reducing manual analysis effort by 60%+.
- Designed a modular agent-based backend with FastAPI for automated responses and scam pattern analysis with sub-second inference latency.

EDUCATION

Manipal University Jaipur, India
Bachelor of Technology in Computer Science Engineering - AIML — 2023 – 2027